MILITARY SPECIFICATION

VALVE, FUEL TANK PRESSURIZATION

This Specification has been approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of fuel tank pressurization valve referenced herein as the '{valve." The valve is actuated by an electrically fired squib.

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-889	Metals, Definition of Dissimilar

DRAWINGS

Naval Sea Systems Command

NAVSEA Dwg 2526708	Valve,Fuel Tank Pressurization, Specification Control Drawing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-1355

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3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the valve are classified as follows:

REQUIREMENT	PARAGRAPH
General requirements	3.2
Performance requirements and product characteristics	3.3
Environmental requirements	3.4
Operational service life	3.5

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or purchase order, a preproduction sample of the valve shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the contractors production methods are capable of yeilding items that comply with the technical requirements of the contract.

3.2.2 Materials

3.2.2.1 Materials and treatments of materials. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the valve shall be in accordance with the applicable documents listed on NAVSEA Dwg 2526708.

3.2.2.2 Metals. Metals shall be of the corrosion-resistant type or shall be adequately protected to resist corrosion during the normal storage and service life of the valve. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact.

3.2.3 Definitions. Definitions of terms appear in Section 6.

3.3 Performance requirements and product characteristics. Unless otherwise specified, the valves shall satisfy the following performance requirements when tested as specified in Section 4:

3.3.1 Leakage at proof pressure. The valve shall be capable of withstanding a maximum internal pressure of 4500 pounds per square inch gage (psig) without leaking and without sustaining permanent distortion.

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3.3.2 Leakage at operating pressure, after firing. After firing, with the squib remaining in the valve body, the valve shall be capable of withstanding an internal operating pressure of 4500 psig when pressure is applied to the inlet and outlet ports.

3.3.3 Squib bridge wire resistance, before firing. The electrical circuit of the bridge wire of all squibs, whether separate or assembled into the valve, shall be continuous and shall have a total resistance of not less than 0.6 ohm nor more than 1.2 ohms, before firing, when measured at a current of 0.01 ampere maximum.

3.3.4 Insulation resistance, valve, before firing. The insulation resistance between the pins of the valve electrical connector and valve case shall be 2 megohms minimum before firing when measurement is made using a potential of 500 volts root-mean-square (rms) at 60 Hertz (Hz).

3.3.5 Insulation resistance, squib, before firing. The insulation resistance between the leads of the squib bridge wires and the squib case shall be 2 megohms minimum before firing when measurement is made using a potential of 500 volts rms at 60 Hz.

3.3.6 Auto-ignition

3.3.6.1 Low temperature. The squibs, whether separate or assembled into the valve body, shall be capable of withstanding a temperature of +170°F for 2 hours and shall not self-ignite. Thereafter, the squib or valve shall meet the operating requirements of 3.3.8.

3.3.6.2 High temperature. The squibs, whether separate or assembled into the valve body, shall be capable of withstanding a temperature of +200°F for 8 hours and shall not self-ignite. Thereafter, the squib, or valve, is not required to meet the operating requirements.

3.3.7 No-fire current. The squib, whether separate or assembled into the valve body, shall be capable of having a current of 0.2 ampere dc passed through its bridge wire for 5 minutes minimum without igniting. Thereafter, the squib shall be capable of meeting the operating requirements of 3.3.8.

3.3.8 Firing current. The squib, whether separate or assembled into the valve body, shall fire when subjected to a current of 2 amperes dc for 5 milliseconds.

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* 3.3.9 Flow, after firing. Upon actuation, the ram shall completely sever the integral metal diaphragm permitting full flow through the valve. The opening shall be capable of flowing a minimum of 5 standard cubic feet per minute of dry gaseous nitrogen with 300 ±5 psig applied to the inlet at 70 ±5°F.

3.3.10 Configuration

3.3.10.1 Dimensions. Dimensions of the valve shall be in accordance with the NAVSEA Dwg 2526708 and all documents listed thereon.

3.4 Enviromental requirements. Unless otherwise specified, the valve shall be capable of conforming to the performance requirements and product characteristics of the specification after being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and nonoperating see 6.3.1, 6.3.2, and 6.3.3.

3.4.1 Vibration, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table I. The valve shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45° F to +140° F. Vibration is defined in 6.3.4.

TABLE I

VIBRATION SCHEDULE

Double Amplitude (in. ± 10%) Minimum	Acceleration (g's ± 10%) Minimum	Frequency (Hz)
0.2	---	5 - 22
---	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

3.4.2 Acceleration, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready conditiion, to the accelerations defined in Table II for a duration of 1 minute for each magnitude. Acceleration is defined in 6.3.5.

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TABLE II

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table III. Shock is defined in 6.3.6.

TABLE III

SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impack Axis	Direction of Deceleration Force	Number Of Impacks
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity

3.4.4.1 Nonoperating

a. Temperature. The valve shall be capable of meeting the performance requirements subsequent to withstanding, in the nonoperating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

b. Humidity. The valve shall be capable of meeting the performance requirements subsequent to withstanding, in the nonoperating condition, an environment of at least 95 percent relative humidity at +110°F for a minimum period of 8 hours. Humidity is defined in 6.3.8.

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3.4.4.2 Ready, operating. The valve, immediately after being conditioned for 1 hour at any temperature between 0° F and +140°F, shall be capable of meeting the performance requirements at any environmental temperature between 0° F and +140° F.

3.5 Operational service life. The valve shall have a service life of one duty cycle.

3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips and loose or spattered foreigh materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of te contract relative to workmanship insofar as not specifically covered by applicable specification.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the valve shall be classified as follows:

a. Preproduction inspection (paragraph 4.4)

b. Quality conformance inspection (paragraph 4.5)

c. Periodic production inspection (paragraph 4.6)

4.3 Sampling

* 4.3.1 Preproduction inspection sample. A preproduction sample of 12 valves, manufactured in accordance with 3.2, shall be subjected to the preproduction tests listed in Table IV and detailed in 4.4 through 4.4.4.14 at an activity designated by the procuring activity. In addition a quantity of 25 squibs shall be subjected to the tests detailed in 4.5.7 through 4.5.7.5. Acceptance of the preproduction sample shall be based on no defects in the sample. See 4.4.6. Further production of the valve by the supplier prior to approval of the preproduction sample, shall be at the supplier's risk.

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* 4.3.2 Quality conformance inspecion sample. The sampling plans'" and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of the lot of valves inspected as specified in 4.5.6.4 and 4.5.6.5. In addition, a quantity of sample squibs equal to 10 percent of the number of squibs assembled into the acceptance lot of valves shall be submitted for acceptance testing as specified in 4.5.7 through 4.5.7.5.

4.3.3 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the quality conformance inspections of “4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 3 units from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to step 2.

Step 2. Randomly select 3 units from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity, for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity, for approval, the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

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4.4 Preproduction inspection

* 4.4.1 Test and examinations. The preproduction sample of 12 valves shall be inspected-in accordance with Table IV, with 2 valves being tested in accordance with each of the 6 test sequences, A through F. In addition, the sample quantity of 25 squibs shall be subjected to the tests detailed in 4.5.7 through 4.5.7.5.

TABLE IV

PREPRODUCTION TEST SEQUENCE

Test	Par. Ref.	Test Sequence					
		A	B	C	D	E	F
Inspection, Prior to Firing	4.4.4.1	1	1	1	1	1	1
Auto-Ignition, Low Temperature	4.4.4.2	2		2	2	2	2
Auto-Ignition, High Temperature	4.4.4.3		2				
Vibration, Ready Condition	4.4.4.4	3					
Acceleration, Ready Condition	4.4.4.5			3			
Shock, Ready Condition	4.4.4.6				3		
Temperature and Humidity	4.4.4.7			4	4		
Low Temperature Operating	4.4.4.8			5			
High Temperature Operating	4.4.4.9				5		
No-Fire Current	4.4.4.10	4					
Firing Current	4.4.4.11	5				4	4
Visual Inspection, After Firing	4.4.4.12	6		6	6	5	5
Flow, After Firing	4.4.4.13	7		7	7	6	6
Leakage at Operating Pressure, Inlet and Outlet Ports, After Firing	4.4.4.14	8		8	8	7	7

4.4 Test conditions. Unless otherwise specified, the valves shall be subjected to the tests in 4.4.1 under the following conditions:

a. Temperature: Room ambient +32° F to +98° F
b. Altitude: Normal ground atmospheric pressure
c. Vibration: None
d. Humidity: Room ambient

4.4.3 Test equipment and setup. In addition to the items listed in 4.5.5., test equipment is required which provides the following characteristics:

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4.4.3.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

a. Frequency range, logarithmic sweep: 5 to 500 Hertz (Hz)
b. Test specimen: 2 pounds
c. Acceleration and displacement (ready condition):

5 to 22 Hz	0.2 inch
22 to 40 Hz	3.0 gravitational units (g's)
40 to 74 Hz	0.036 inch
74 to 500Hz	10 g's

d. Acceleration (packed and packaged):

5 to 60 Hz	1 g

e. Automatic cycling time range:	0 to 15 minutes (optional)
f. Temperature chamber range:	-45° F to +140° F, accuracy to ±5° F
g. Interior size:	6 inches x 4 inches x 3 inches minimum

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with a 2 pound test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting a 2 pound test specimen and shall provide ranges of:

a. 300 g's for 0.5 millisecond (ms) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ms to 20.0 ms

4.4.3.4. Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics:

a. Temperature:	-45°F to +250° F; with an accuracy to ±5° F
b. Relative humidity:	95 percent minimum at +110° F; with an accuracy to ±5° F
c. Interior size:	6 inches x 4 inches x 3 inches minimum

This equipment may be used for both nonoperating and ready/operating temperature (and humidity) tests.

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4.4.4 Test procedures

4.4.4.1 Inspection. The valve shall be subjected to the inspections, prior to firing, detailed in 4.5.6.1 through 4.5.6.4.

* 4.4.4.2 Auto-ignition, low temperature. The valve (or sample squib) shall be heated in air to a temperature of +170° F ±2° F and maintained at that temperature of 2 hours. The valve (or sample squib) shall not fire. Thereafter, the valve (or sample squib) shall be capable of meeting the requirements of 4.4.4.11.

4.4.4.3 Auto-ignition, high temperature. The valve shall be heated in air to a temperature of +200° F ±10° F and maintained at. that temperature for 8 hours. The valve shall not fire. Thereafter, the valve shall not be required to meet the operating characteristics:

4,4.4.4 Vibration, ready condition. The valve, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 Hz to 500 Hz. The valve shall then be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table V. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -450 F ±50 F, room temperature and +140° F ±5° F respectively. For cycling, a logarithmid frequency sweep rate shall be used if avaiable; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes defined in Table VI.

TABLE V

VIBRATION TEST SCHEDULE

Number of Resonances	0	1	2	3	4
Total Vibration Time at Resonance *	--	30 min	1 hr	1-½ hr	2 hr
Cycling Time	3 hr	2-½ hr	2 hr	1-½ hr	1 hr

* 30 minutes each resonance

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TABLE VI

VIBRATIONS SCHEDULE

Frequency (Hz)	5-22	22-40	40-74	74-500
Double Amplitude (in. ± 10%) Minimum	0.2	---	0.036	---
Acceleration (g's ± 10%) Minimum	---	3	---	10

4.4.4.5 Acceleration, ready condition. The valve, in the ready condition, shall be subjected to the accelerations defined in Table VII after being secured to an acceleration equipment (see Figure 1). Acceleration shall be measured at the approximate geometric center of the valve.

TABLE VII

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (see Figure 1)	Time
25	Longitudinal axis, Fwd	1 Minute
35	Longirudinal axis, Aft	1 Minute

4.4.4.6 Shock, ready condition. The valve, in the ready condition, shall be attached to a shock test machine (see Figure 1). Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the valve shall be performed in accordance with Table VIII in the sequence shown.

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TABLE VIII

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impack Axis	Direction of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

4.4.4.7 Temperature and humidity, nonoperating condition. Four preproduction valves shall be subjected, in the nonoperating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F +0° -5°F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45° F +0° -5° F within 2 hours maximum. The specimen shall therafter remain in the chamb for 4 hours). The specimen shall then be subjected to an environmental temperature of +140° F +5° F -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110° F +5° -0° F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Thereafter, 2 valves shall be operated in accordance with 4.4.4.8 and 2 valves shall be operated in accordance with 4.4.4.9. Prior to operation the valves may be wiped dry. Drying by an air blast or oven is not permitted.

* 4.4.4.8 Low temperature operating. Before testing, the valves installed in its test chamber in the ready condition, shall be conditioned at a temperature of 0° F ±5° F for a period of 1 hour. The valve shall then be operated in accordance with and meet the requirements of 4.4.4.11 at an environmental temperature of +32° F ± 5° F. The valve shall thereafter meet the requirements of 4.4.4.12 through 4.4.4.14.

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* 4.4.4.9 High temperature operating. Before testing, the valve, installed in its test chamber in the ready condition, shall be conditioned at a temperature of +140° F ±5° F for a period of 1 hour. The valve shall then be operated in accordance with and meet the requirements of 4.4.4.11 at an environmental temperature of +140°F ±5° F. The valve shall thereafter meet the requirements of 4.4.4.12 through 4.4.4.14.

4.4.4.10 No-fire current. The valve (or sample squib) shall be subjected to a current of 0.2 ±0.02 ampere maximum for a period of 5 minutes minimum, in accordance with Figure 2, and shall not fire. Thereafter, the valve (or sample squib) shall be capable of meeting the requirements of 4.4.4,11.

4.4.4.11 Firing current. The valve (or sample squib) shall fire when subjected to a current of 2.0 ±0.2 ampere dc for 5 ±0.5 milliseconds in accordance with Figure 2.

4.4.4.12 Visual inspection, after firing. After firing, the valve case shall be inspected and shall show no evidence of permanent distortion.

* 4.4.4.13 Flow, after firing. Upon actuation, the ram shall completely sever the integral metal diaphragm permitting full flow through the valve. The flow through the actuated valve shall be at least 5 standard cubic feet per minute of dry gaseous nitrogen with 300 ±5 psig applied to the inlet at 70 ±5°F.

4.4.4.14 Leakage at operating pressure, inlet and outlet ports, after firing. With the squib remaining in the valve body after firing, the valve shall be subjected to a pressure of 4500 ±200 psig, using dry inert gas. Pressure shall be applied to the inlet and outlet ports for a period of 5 minutes with the valve submerged in water. There shall be no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion.

* 4.4.5 Test data sheets. All test data shall be recorded on a date sheet by the supplier and maintained as part of the inspection records. In addition to the data requirements of 4.5.8, the data sheets shall contain. but are not limited to, the following information:

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Preproduction Test	Valve Serial Number 1	2	3	4	5	6	7	8	9	10	11	12
Inspection, Prior to Firing (Data per 4.5.8)												
Auto-Ignition, Low Temperature												
Passed												
Failed												
Auto-Ignition, High Temperature												
Passed												
Failed												
Low Temperature, Operating												
Passed												
Failed												
High Temperature, Operating												
Passed												
Failed												
No-Fire Current												
Passed												
Failed												
Firing Current												
Passed												
Failed												
Flow, After Firing, SCFM												

* 4.4.6 Determination of defect or failure. Failure of a valve to successfully pass any of the tests in 4.4 through 4.4.4.14 Or failure of a sample- squib to-pass any of the tests in 4.5.7 through 4.5.7.5 shall be cause for rejection of the preproduction sample.

* 4.5. Quality conformance inspection. Valves produced under the contract or purchase order shall be subjected to the following tests and examinations. A quantity of additional sample squibs shall be provided-which is equal to 10 percent of the number of squibs to be assembled into the valves in the lot being submitted for acceptance testing.

4.5.1 Tests and examinations

4.5.1.1 Valve

a. Leakage at proof pressure
b. Bridge wire resistance
c. Insulation resistance
d. Configuration
e. Operational test

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4.5.1.2 Sample squibs

a. Bridge wire resistance, prior to firing
b. Insulation resistance, prior to firing
c. Auto-ignition, low temperature
d. No-fire current
e. Firing current

4.5.2 Each valve in the lot shall be subjected to the tests detailed in 4.5.6.1 through 4.5,6.3.

4.5.3 A sample of valves shall be selected using-the plans and procedures set forth in MIL-STD-105 Inspection Level 11, and subjected to the additional tests and examination in 4.5.6.4 and 4.5.6.5. An AQL of 1.0 shall be used.

4.5.4 Test conditions. Unless otherwise specified, the valve shall be subjected to the acceptance tests under the following conditions:

a. Temperature:	Room ambient +32° F to +98° F
b. Altitude:	Normal ground atmospheric pressure
c. Vibration:	None
d. Humidity:	Room ambient

4.5.5 Test equipment and setup. The test equipment setup shall be in accordance with Figure 2.

a. Inert gas supply:	O to 4700 psig minimum
b. Pressure gage:	O to 5000 psig, accuracy to ±2 percent, full scale
c. Power supply:	(1) Constant current, presettable pulse amplitude and duration, 2 amperes, 5 ±0.5 ms pulses . into loads of 0.6 to 1.2 ohms (2) Constant current, 0.2 ±0.02 ampere, pulses 5-minute minimum into loads-of 0.6 to 1.2 ohms.
d. Megger:	500 ±5O volts rms, 60 Hz across a load of 2 megohms minimum
e. Resistance measuring device:	Accuracy to ±3 percent, readout. resolution 0.1 ohms minimum, current 10 milliamperes maximum.

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4.5.6 Test procedures, valve

4.5.6.1 Leakage at proof pressure. Prior to assembly of a squib into the valve body, the valve shall be subjected to a pressure of 4500 ±200 psig using dry inert gas. Pressure shall be applied to the outlet port for 5 minutes. The valve shall be submerged in water and shall show no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion. This test shall be repeated with pressure applied for 5 minutes simultaneously to the inlet and outlet ports. The valve shall be re-examined for evidence of permanent distortion.

4.5.6.2 Bridge wire resistance. The resistance of the squib bridge wire shall be determined. A current of 0.01 ampere maximum shall be applied to the pins of the valve electrical connector. Resistance shall be not less than 0.6 ohm nor more than 1.2 ohms.

4.5.6.3 Insulation resistance. The valve shall be subjected to a potential of 500 ±5O volts rms, 60 Hz, for 1 second minimum, applied between each electrical connector pin and the case. The pins may be connected together to test simultaneously if desired. Resistance shall be 2 megohms minimum.

* 4.5.6.4 Configuration. Ref: NAVSEA Dwg 2526708. All characteristics are MINOR.

a. Materials. Materials and finish shall be in accordance with the drawing.
b. Dimensions.

.750 +0.005 -0.010 hex dim
½-20 UNF-3A major dia MIN
½-20 UNF-3A pitch dia
3/8-24 UNF-3B minor dia MAX
3/8-24 UNF-3B pitch dia

c. Marking. The marking shall be in accordance with Dwg 2526708.

* 4.5.6.5 Operational test. The sample valves shall be subjected to the firing current test detailed in 4.4.4.11 and the visual inspection detailed in 4.4.4.12.

4.5.7 Test procedures, sample squibs

4.5.7.1 Bridge wire resistance, prior to firing. The resistance of the squib bridge wire shall be determined. A current of 0.01 ampere maximum shall be applied to the bridge lead wires. Resistance shall be not less than 0.6 ohm nor more than 1.2 ohms,

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4.5.7.2 Insulation resistance, prior to firing. The sample squib shall be subjected to a potential of 500 ±50 volts rms, 60 Hz, for 1 second minimum, applied between the bridge lead wires (connected together) and the case of the sample squib. Resistance shall be 2 megohms minimum.

4.5.7.3 Auto-ingition, low temperature. The sample squibs shall be subjected to the test detailed in 4.4.4.2.

* 4.5.7.4 No-fire current. The sample squibs shall be subjected to the test detailed in 4.4.4.10.

* 4.5.7.5 Firing current. The sample squibs shall be subjected to the test detailed in 4.4.4.11.

4.5.8 Test data sheets. All test data shall be recorded on date sheets by the supplier. Data sheets shall contain, but are not limited to, the following information;

4.5.8.1 Valve

Date:

Valve Serial Number	Proof Pressure		Bridge Wire Resistance	Insulation Resistance	Firing Test	
	Passed	Failed	Ohms	Ohms	Passed	Failed

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4.5.8.2 Sample squibs

	Sample Squib Serial Number											
	1	2	3	4	5	6	7	8	9	10	11	12
Bridge Wire Resistance - ohms												
Insulation Resistance - ohms												
Auto-Ignition, Low Temperature												
Passed												
Failed												
No-Fire Current												
Passed												
Failed												
Firing Current												
Passed												
Failed												

4.5.9 Determination of defect or failure. Failure of a valve to pass any of the tests in 4.5.6.1 through 4.5.6.3 shall be cause for rejection of that valve. A defective is a unit of product which contains one or more defects. Failure of a valve or sample squib to successfully pass the tests in 4.5.6.4 or 4.5.6.5. shall constitute a defect. Failure of a sample squib to pass any of the tests in 4.5.7.1 through 4.5.7.5 shall be cause for rejection of the acceptance lot.

* 4.6 Periodic production inspection. Periodic production samples shall be selected in accordance with 4.3.3. The samples shall be tested in accordance with the following plan:

a. The sample of 3 units from the first group of 30 units will be called the first sample and will undergo the tests of Table IX for Sample 1. The sequence of tests will be as shown in Table IX.

b. The sample of 3 units from the second and third groups of 30 units-will be called the second and third samples and will undergo the tests of Table IX for sample 2 and 3 respectively. The sequence of tests will be as shown in Table IX.

c. The next sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This sample will be called the fourth sample and will undergo the tests of Table IX for sample 4, etc.

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TABLE IX

PERIODIC TEST SCHEDULE

Tests	Sample and Test Sequence								
	1, 4, etc.			2, 5, etc.			3, 6, etc.		
Inspection, Prior to Firing	1	1	1	1	1	1	1	1	1
Vibration, Ready Condition	2*						2*		
Acceleration, Ready Condition		2*						2*	
Shock, Ready Condintion				2*					2*
Temperature and Humidity					2		3		
No-Fire Current	3				3	2			3
Auto-Ignition, Low Temperature		3						3	
Auto-Ignition, High Temperature	4				4				
Low Temperature, Operating		4				3		4	
High Temperature, Operating			2				4		
Firing Current		5	3	3		4	5	5	4
Visual Inspection, After Firing		6	4	4		5	6	6	5
Flow, After Firing		7	5			6	7	7	
Leakage at Operating Pressure, Inlet and Outlet Ports, After Firing		8	6			7	8	8	6

* Visual Check After Testing

4.6.1 Test conditions. Unless otherwise specified the valve shall be subjected to the periodic tests under the following conditions:

a. Temperature: Room ambient +32° F to +98° F
b. Altitude: Normal ground atmospheric pressure
c. Vibration: None
d. Humidity: Room ambient

4.6.2 Test equipment and setup. The test equipment and setup for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test procedures

4.6.3.1 Inspection, prior to firing. Inspection of the valve, prior to firing, shall be performed in accordance with 4.5.6.1 through 4.5.6.4.

4.6.3.2 Vibration, ready condition. Vibration testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.4.

4.6.3.3 Acceleration, ready condition. Acceleration testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.5.

4.6.3.4 Shock, ready condition. Shock testing of the valve in the ready condition, shall be performed in accordance with 4.4.4.6.

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4.6.3.5 Temperature and humidity, nonoperation condition. Temperature and humidity testing of the valve, in the nonoperating condition, shall be performed in accordance with 4.4.4.7.

* 4.6.3.6 No-fire current. The valve shall be subjected to the no-fire current test in accordance with 4.4.4.10.

4.6.3.7 Auto-ignition, low temperature. The valve shall be subjected to the auto-ignition, low temperature, test in accordance with 4.4.4.2.

4.6.3.8 Auto-ignition, high temperature. The valve shall be subjected to the auto-ignition, high temperature, test in accordance with 4.4.4.3.

4.6.3.9 Low temperature operating. The valve shall be subjected to the low temperature operating test in accordance with 4.4.4.8.

4.6.3.10 High temperature operating. The valve shall be subjected to the high temperature operating test in accordance with 4.4.4.9.

* 4.6.3.11 Firing current. The valve shall be subjected to the firing current test in accordance with 4.4.4.11.

* 4.6.3.112 Leakage at operating pressure, inlet and outlet ports, after firing. The valve shall be subjected to the post firing leakage test (inlet and outlet ports) in accordance with 4.4.4.14.

4.6.3.13 Flow, after firing. The valve shall be subjected to the flow test in accordance with 4.4.4.13.

* 4.6.3.14 Visual inspection, after firing. The valve, after firing, shall be subjected to visual inspection in accordance with 4.4.4.12.

4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as a part of the inspection record.

4.6.5 Determination of defect or failure. Failure of the valve to successfully pass the tests in 4.6.3.1 through 4.6.3.14 shall be cause for withholding acceptance of all units on hand. Corrective action shall be taken in accordance with 4.3.3.

* 5. PREPARATION FOR DELIVERY
(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. The valve is intended for use in a torpedo to pressurize the torpedo engine fuel tank.

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* 6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements

a. Title, number, and date of this specification
b. Assigned activity for preproduction tests
c. Any exceptions to this specification
d. Inspection data required,
e. Hazard classification
f. That the safety precaution requirementsof the Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material, DOD 4145.26M, are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP5, should be made applicable.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs herein:

Paragraph	Data Requirement	Applicable DIDS
4.4.5	Test data	*
4.5.8	Test data	*
4.6.4	Test data	*

*DIDS (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Definitions

6.3.1 Ready. The condition of the valve when fully assembled, prior to application of firing current.

6.3.2 Operating. The condition of the valve when fully assembled. after application of firing current.

6.3.3 Nonoperating. The condition of the valve when stored. transported, or handled. Tests on the valve, in the nonoperating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in Hertz (Hz).

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6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in gravitational units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a buildup to 90 percent of peak force in less than 0.010 second. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds, and shock wave form equivalent to one-half sine wave. Duration is defined as the interval between the point of zero amplitude, from which the shock wave form builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the valve is exposed, is expressed in degrees Fahrenheit (F).

6.3.8 Humidity. Humidity is the relative humidity of the environment to which the valve is exposed, expressed in percent.

6.3.9 Valve. This term refers to the complete explosive valve unit, which includes the independent, electrically fired squib assembly contained in the valve body.

6.3.10 Sample squib. This term refers to serarate sample quibs submitted with each lot of valves for preproduction or quality conformance testing.

6.4 Suitability. Any valve which has been subjected to a non-firing current in excess of the continuity test current of 0.200 ampere shall be considered unsuitable for service use and shall be expended.

6.5 Weight. The weight of the valve shall not exceed 0.5 pound.

* 6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenince only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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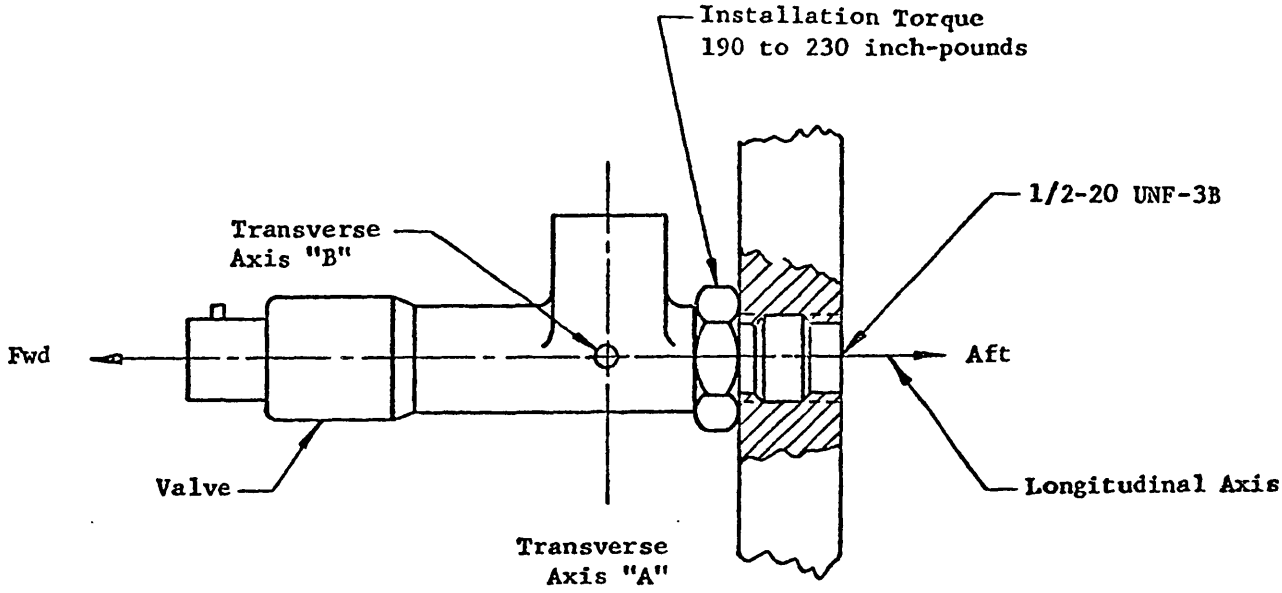


Figure 1. Test Fixture

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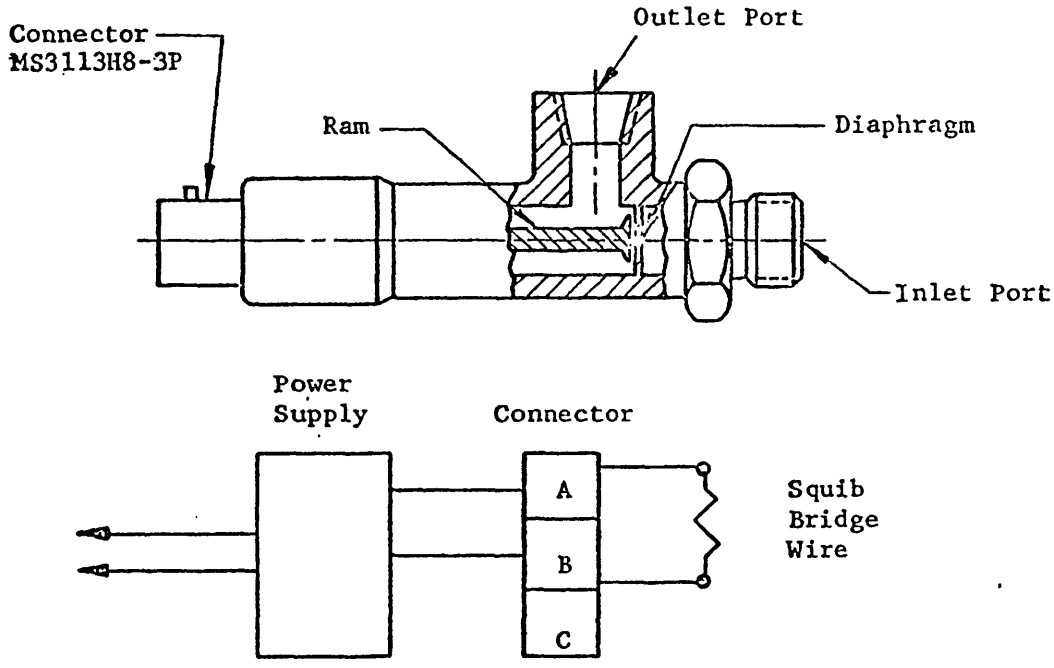


Figure 2. Valve Details and Schematic Diagram

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DOCUMENT IDENTIFIER AND TITLE
MIL-V-82444A(OS) "Valve, Fuel Tank Pressurization"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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